

Introduction to Data Pre-processing

Data Pre-processing is the process of cleaning, transforming, reducing, and organizing raw data into a suitable format for analysis, machine learning, and Business Intelligence applications.

Raw data collected from different sources is often:

- Incomplete
- Noisy
- Inconsistent
- Redundant

Poor quality data reduces model accuracy and affects business decisions.

Data preprocessing improves:

- Data quality
- Accuracy
- Reliability
- Processing efficiency

Objectives of Data Pre-processing

1. Improve data quality
2. Remove errors and inconsistencies
3. Reduce data complexity
4. Prepare data for analytics and ML
5. Improve model performance

Data Pre-processing Workflow

Raw Data
↓
Data Cleaning
↓
Transformation
↓
Reduction
↓
Prepared Data

1. Data Validation

Data Validation is the process of checking whether data is correct, complete, and meaningful before processing.

Validation ensures that invalid or incorrect data does not enter the system.

Types of Validation

A. Range Validation

Checks whether values lie within a valid range.

Example

Age should be between 1 and 120.

B. Format Validation

Checks proper data format.

Example

Email format validation.

C. Type Validation

Checks correct datatype.

Example

Phone number should contain numeric values.

Advantages

1. Improves data accuracy.
2. Prevents incorrect processing.
3. Reduces errors.

Disadvantages

1. Additional processing time.
2. Complex validation rules.

Applications

- Database systems
- Web forms
- Machine learning preprocessing

2. Incomplete Data

Incomplete Data refers to datasets containing missing or unavailable values.

Causes of Incomplete Data

1. Data entry errors
2. Sensor failures
3. Missing survey responses
4. System issues

Problems Caused

1. Reduced model accuracy
2. Incorrect analysis
3. Biased predictions

Handling Techniques

A. Remove Missing Records

Delete incomplete rows.

B. Fill Missing Values

Replace missing values using:

- Mean
- Median
- Mode

C. Predict Missing Values

Use machine learning techniques.

Advantages of Handling Incomplete Data

1. Better data quality.
2. Improved analytics accuracy.

3. Data Affected by Noise

Noisy Data contains random errors, inconsistencies, or unwanted information.

Causes of Noise

1. Sensor errors
2. Human mistakes
3. Transmission errors
4. Duplicate entries

Examples

- Incorrect age values
- Typing mistakes
- Corrupted records

Noise Handling Techniques

A. Smoothing

Removes random variations.

B. Filtering

Eliminates unwanted data.

C. Clustering

Detects outliers and abnormal data.

Advantages

1. Improved data quality.
2. Better prediction accuracy.

Disadvantages

1. Risk of losing important data.

Applications

- Sensor systems
- Medical datasets
- Financial analytics

4. Data Transformation

Data Transformation converts data into a suitable format for analysis and machine learning.

Objectives

1. Improve consistency
2. Normalize data
3. Prepare features for models

Types of Data Transformation

1. Standardization
2. Normalization
3. Aggregation
4. Encoding

Advantages

1. Better model performance.
2. Improved data compatibility.

5. Standardization

Standardization transforms data so that features have similar scale with mean 0 and standard deviation 1.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

$$x$$

$$\mu$$

$$\sigma$$

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

Where:

- X = Original value
- μ = Mean
- σ = Standard deviation

Importance

Different scales affect machine learning algorithms.

Example

Age values:

18, 25, 40

Income values:

10000, 50000, 90000

Standardization converts them into comparable scales.

Advantages

1. Improves ML performance.
2. Faster convergence.

Disadvantages

1. Sensitive to outliers.

Applications

- Machine learning
- Neural networks
- Statistical analysis

6. Feature Extraction

Feature Extraction is the process of selecting or creating important attributes from raw data.

Large datasets may contain unnecessary features.

Feature extraction reduces complexity while preserving useful information.

Examples

- Extracting edges from images
- Extracting keywords from text

Advantages

1. Reduces dimensionality.
2. Improves model accuracy.
3. Faster computation.

Disadvantages

1. Possible information loss.

Applications

- Image processing
- NLP
- Pattern recognition

Conclusion

Feature extraction identifies meaningful information for efficient analytics.

7. Data Reduction

Data Reduction reduces dataset size while maintaining important information.

Objectives

1. Reduce storage space
2. Improve processing speed
3. Simplify analysis

Types of Data Reduction

1. Sampling
2. Feature selection
3. PCA

Advantages

1. Faster processing.
2. Lower storage cost.

Disadvantages

1. Information loss possible.

8. Sampling

Sampling selects a smaller subset of data from a larger dataset.

Types of Sampling

A. Random Sampling

Records selected randomly.

B. Stratified Sampling

Data divided into groups before sampling.

C. Systematic Sampling

Records selected at fixed intervals.

Advantages

1. Reduces processing time.
2. Lower computational cost.

Disadvantages

1. Sampling bias possible.

Applications

- Surveys
- Big data analytics
- Statistical analysis

9. Feature Selection

Feature Selection identifies the most relevant features for model training. Irrelevant features reduce performance and increase complexity.

Methods of Feature Selection

A. Filter Method

Uses statistical techniques.

B. Wrapper Method

Tests different feature subsets.

C. Embedded Method

Feature selection occurs during training.

Advantages

1. Improved model accuracy.
2. Reduced overfitting.
3. Faster computation.

Disadvantages

1. Important features may be removed accidentally.

Applications

- Machine learning
- Predictive analytics
- Classification systems

10. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms

high-dimensional data into fewer principal components.

PCA reduces complexity while preserving maximum variance.

Working of PCA

Step 1:

Standardize data.

Step 2:

Compute covariance matrix.

Step 3:

Calculate eigenvectors and eigenvalues.

Step 4:

Select principal components.

Advantages

1. Reduces dimensionality.
2. Faster processing.
3. Removes redundancy.

Disadvantages

1. Reduced interpretability.
2. Information loss possible.

Applications

- Image compression
- Pattern recognition
- Big data analytics

11. Data Discretization

Data Discretization converts continuous data into discrete intervals or categories.

Example

Age:

0–18 → Child

19–60 → Adult

60+ → Senior Citizen

Methods of Discretization

A. Equal Width Method

Intervals have equal size.

B. Equal Frequency Method

Intervals contain equal number of records.

Advantages

1. Simplifies analysis.
2. Improves model interpretability.

Disadvantages

1. Loss of detailed information.

Applications

- Data mining
- Classification systems
- Statistical analysis

Introduction to Data Exploration

Data Exploration is the process of analyzing datasets to understand their structure, patterns, trends, relationships, and characteristics before applying machine learning or analytics techniques.

Raw data may contain:

- Missing values
- Outliers
- Noise
- Inconsistencies

Data exploration helps in:

- Understanding data distribution
- Detecting anomalies
- Identifying trends
- Selecting preprocessing methods

Objectives of Data Exploration

1. Understand dataset characteristics
2. Detect outliers and anomalies
3. Analyze data distribution
4. Identify patterns and trends
5. Improve decision-making and model accuracy

Types of Data Exploration

1. Univariate Analysis
2. Bivariate Analysis
3. Multivariate Analysis

Univariate Analysis

Definition

Univariate Analysis analyzes only one variable or attribute at a time.

It helps understand:

- Distribution
- Central tendency
- Variability
- Outliers

Objectives of Univariate Analysis

1. Summarize data characteristics
2. Detect unusual values
3. Understand data distribution
4. Prepare data for modeling

Univariate Analysis Workflow

Single Variable



Graphical Analysis



Statistical Analysis



Insights and Patterns

1. Graphical Analysis of Categorical Attributes

Graphical analysis of categorical attributes represents categorical data visually to understand frequency and distribution.

Categorical attributes contain labels or categories instead of numeric values.

Examples

- Gender
- Department
- Product category

Common Graphs for Categorical Data

A. Bar Chart

A Bar Chart represents categorical data using rectangular bars.

Example

Department:

IT 

HR 

Sales 

Advantages

1. Easy comparison.
2. Simple visualization.
3. Easy interpretation.

Applications

- Product analysis
- Survey results
- Sales categories

B. Pie Chart

Pie Chart represents categories as portions of a circle.

Example

Market share distribution.

Advantages

1. Shows percentage contribution.
2. Easy visual understanding.

Disadvantages

1. Difficult with many categories.

Applications

- Budget analysis
- Customer segmentation

C. Count Plot

Displays frequency count of each category.

Applications

- Frequency analysis
- Classification datasets

2. Graphical Analysis of Numerical Attributes

Graphical analysis of numerical attributes visually represents numerical data distributions and patterns.

Numerical attributes contain measurable values.

Examples

- Salary
- Age

- Marks
- Temperature

Common Graphs for Numerical Data

A. Histogram

Histogram displays frequency distribution of numerical data using continuous bars.

Example

Marks distribution among students.

Advantages

1. Shows distribution clearly.
2. Detects skewness and spread.

Applications

- Statistical analysis
- Performance analysis

B. Box Plot

Box Plot represents distribution using quartiles and highlights outliers.

Components

- Minimum value
- First quartile (Q1)
- Median
- Third quartile (Q3)
- Maximum value

Advantages

1. Detects outliers easily.
2. Shows spread and skewness.

Applications

- Financial analysis
- Data preprocessing

C. Density Plot

Density Plot shows smooth probability distribution of data.

Applications

- Statistical analysis
- Machine learning preprocessing

D. Stem-and-Leaf Plot

Displays numerical data while preserving actual values.

Applications

- Small dataset analysis

3. Measures of Central Tendency for Numerical Attributes

Measures of Central Tendency describe the central or typical value of a dataset.

Types of Central Tendency

1. Mean
2. Median
3. Mode

A. Mean

Mean is the average of all values.

Formula

$$\bar{x} = \frac{\sum x}{n}$$

Where:

- $\sum x$ = Sum of values
- n = Number of observations

Example

Values:

10, 20, 30

Mean:

$$(10 + 20 + 30)/3 = 20$$

Advantages

1. Easy to calculate.
2. Uses all observations.

Disadvantages

1. Sensitive to outliers.

B. Median

Median is the middle value in sorted data.

Example

5, 10, 15

Median = 10

Advantages

1. Not affected by outliers.

Disadvantages

1. Does not use all values.

C. Mode

Mode is the most frequently occurring value.

Example

2, 2, 3, 5

Mode = 2

Advantages

1. Useful for categorical data.

Disadvantages

1. May not exist uniquely.

Comparison of Central Tendency Measures

Measure Best Use		Sensitive to Outliers
Mean	Normally distributed data	Yes
Median	Skewed data	No
Mode	Categorical data	No

4. Measures of Dispersion for Numerical Attributes

Measures of Dispersion describe how spread out data values are from the central value.

Importance

Dispersion helps understand:

- Variability
- Consistency
- Data spread

Types of Dispersion Measures

1. Range
2. Variance
3. Standard Deviation
4. Interquartile Range (IQR)

A. Range

Difference between maximum and minimum values.

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Example

10, 20, 50

$$\text{Range} = 50 - 10 = 40$$

B. Variance

Variance measures average squared deviation from the mean.

Formula

$$\sigma^2 = \frac{\sum(x - \mu)^2}{N}$$

Advantages

1. Measures variability accurately.

Disadvantages

1. Difficult interpretation due to squared units.

C. Standard Deviation

Standard deviation measures average spread of data around the mean.

Formula

$$\sigma = \sqrt{\frac{\sum(x - \mu)^2}{N}}$$

Advantages

1. Widely used.
2. Easy interpretation.

Applications

- Statistical analysis
- Machine learning

D. Interquartile Range (IQR)

IQR measures spread of middle 50% of data.

Formula

$$IQR = Q3 - Q1$$

Advantages

1. Resistant to outliers.

5. Identification of Outliers for Numerical Attributes

Outliers are data values significantly different from other observations.

Causes of Outliers

1. Data entry errors
2. Measurement errors
3. Rare events
4. Fraudulent activities

Problems Caused by Outliers

1. Reduced model accuracy
2. Incorrect analysis
3. Distorted statistical measures

Methods for Outlier Detection

A. Box Plot Method

Uses quartiles and IQR.

Outlier Rule

$$\text{Outlier} < Q1 - 1.5(IQR) \text{ or } \text{Outlier} > Q3 + 1.5(IQR)$$

B. Z-Score Method

Detects values far from mean.

Formula

$$Z = \frac{X - \mu}{\sigma}$$
$$z = \frac{x - \mu}{\sigma} \approx 1.2$$
$$\Phi(z) \approx 88.5\%$$

Rule

If: $|Z| > 3$

Then value is considered an outlier.

C. Scatter Plot

Visual identification of abnormal values.

Advantages

1. Improves data quality.
2. Better model accuracy.
3. Removes abnormal values.

Disadvantages

1. Important rare events may be removed accidentally.

Applications

- Fraud detection
- Medical analysis
- Financial analytics

Introduction to Bivariate Analysis

Bivariate Analysis is the statistical analysis of two variables to study their relationship, association, dependency, or correlation.

Unlike univariate analysis, which studies only one variable, bivariate analysis examines how one variable changes with another.

It helps in:

- Finding relationships
- Detecting correlations
- Understanding dependencies
- Identifying trends and patterns

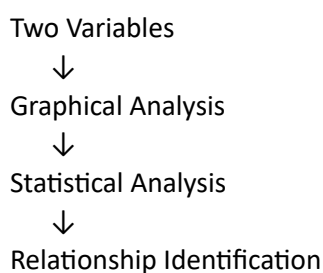
Objectives of Bivariate Analysis

1. Study relationships between variables
2. Detect positive or negative correlations
3. Compare datasets
4. Support predictive analysis
5. Improve decision-making

Types of Variables in Bivariate Analysis

Variable Type	Example
Numerical vs Numerical	Age and Salary
Categorical vs Categorical	Gender and Product Choice
Numerical vs Categorical	Salary by Department

Bivariate Analysis Workflow



1. Graphical Analysis in Bivariate Analysis

Graphical analysis visually represents relationships between two variables using graphs and charts.

Importance

Graphical methods help:

- Detect trends
- Identify clusters
- Observe correlations
- Detect outliers

Common Graphical Methods

A. Scatter Plot

A Scatter Plot displays relationships between two numerical variables using points on a graph.

Example

Relationship between:

- Study hours
- Exam marks

Interpretation

Positive Correlation

As one variable increases, the other also increases.

Negative Correlation

As one variable increases, the other decreases.

No Correlation

No clear relationship between variables.

Advantages

1. Easy relationship visualization.
2. Detects trends and outliers.

Disadvantages

1. Difficult for very large datasets.

Applications

- Business analytics
- Financial analysis
- Machine learning

B. Line Graph

Line Graph shows relationship and trends between variables over time.

Applications

- Stock market analysis
- Sales trend analysis

C. Grouped Bar Chart

Compares two categories visually using grouped bars.

Applications

- Sales comparison by department
- Product category comparison

D. Heatmap

Heatmap uses colors to represent relationships or correlation intensity.

Applications

- Correlation analysis
- Data science visualization

2. Measures of Correlation for Numerical Attributes

Correlation measures the strength and direction of relationship between two numerical variables. Correlation analysis determines whether variables move together or independently.

Types of Correlation

A. Positive Correlation

Both variables increase together.

Example

Experience and salary.

B. Negative Correlation

One variable increases while the other decreases.

Example

Price and demand.

C. Zero Correlation

No relationship between variables.

Correlation Coefficient

Correlation coefficient measures the strength of correlation between variables.

Range

$$-1 \leq r \leq +1$$

Where:

Value of r Meaning

+1 Perfect positive correlation

0 No correlation

-1 Perfect negative correlation

Pearson Correlation Coefficient

Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Advantages

1. Measures linear relationship accurately.
2. Easy interpretation.

Disadvantages

1. Sensitive to outliers.
2. Detects only linear relationships.

Spearman Rank Correlation

Measures correlation based on ranked values instead of actual values.

Formula

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Where:

- d = Difference between ranks
- n = Number of observations

Advantages

1. Works for non-linear relationships.
2. Less affected by outliers.

Applications of Correlation Analysis

- Stock market prediction
- Sales analysis
- Healthcare analytics
- Machine learning

3. Contingency Tables for Categorical Attributes

A Contingency Table is a tabular representation used to analyze relationships between two categorical variables.

It displays frequency counts of combinations of categories.

Example

Gender Likes Product Dislikes Product

Male	40	10
Female	35	15

Components of Contingency Table

- 1. Rows represent one categorical variable
- 2. Columns represent another variable
- 3. Cells contain frequency counts

Purpose of Contingency Tables

- 1. Analyze categorical relationships
- 2. Detect associations
- 3. Compare category frequencies

Chi-Square Test

Chi-Square Test checks whether two categorical variables are independent or associated.

Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- *O*= Observed frequency
- *E*= Expected frequency

Advantages

- 1. Simple categorical analysis.
- 2. Detects associations effectively.

Disadvantages

- 1. Requires sufficient sample size.

Applications

- Survey analysis
- Market research
- Customer behavior analysis

Difference Between Univariate and Bivariate Analysis

Parameter	Univariate Analysis	Bivariate Analysis
Variables Analyzed	One	Two
Purpose	Describe data	Find relationships
Example	Mean of salary	Salary vs Experience
Common Graph	Histogram	Scatter Plot

Advantages of Bivariate Analysis

- 1. Identifies variable relationships
- 2. Supports predictive analysis
- 3. Improves decision-making
- 4. Detects dependencies and trends

Disadvantages of Bivariate Analysis

- 1. Complex interpretation sometimes
- 2. Correlation does not imply causation

Applications of Bivariate Analysis

- 1. Financial analysis
- 2. Marketing analytics
- 3. Healthcare analysis
- 4. Machine learning
- 5. Customer behavior analysis

Introduction to Multivariate Analysis

Multivariate Analysis is the statistical analysis of more than two variables simultaneously to understand relationships, patterns, dependencies, and interactions among variables.

Real-world datasets often contain multiple variables that influence each other.

Example:

- Sales may depend on:
 - Advertising
 - Product price
 - Customer income
 - Season

Multivariate analysis helps study all these variables together.

Objectives of Multivariate Analysis

1. Analyze relationships among multiple variables
2. Identify patterns and trends
3. Reduce data complexity
4. Improve prediction accuracy
5. Support business decision-making

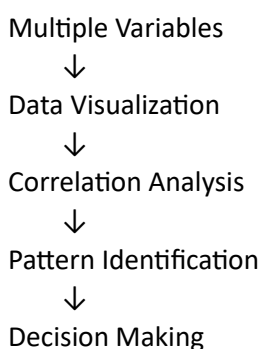
Importance of Multivariate Analysis

1. Handles complex datasets
2. Improves machine learning models
3. Supports predictive analytics
4. Detects hidden relationships
5. Helps in dimensionality reduction

Types of Variables in Multivariate Analysis

Type	Example
Multiple Numerical Variables	Age, Salary, Experience
Numerical + Categorical	Salary, Department
Multiple Categorical Variables	Gender, Product Type, Region

Multivariate Analysis Workflow



1. Graphical Analysis in Multivariate Analysis

Graphical Analysis in Multivariate Analysis visually represents relationships among multiple variables using advanced visualization techniques.

Importance

Graphical methods help:

- Identify hidden patterns

- Detect relationships
- Understand clusters
- Analyze trends

Common Graphical Techniques

A. Scatter Plot Matrix

A Scatter Plot Matrix displays pairwise relationships between multiple numerical variables.

Example

Relationship among:

- Age
- Salary
- Experience
- Education

Advantages

1. Shows multiple relationships together.
2. Detects correlations and trends.

Disadvantages

1. Complex for very large datasets.

Applications

- Machine learning
- Statistical analysis
- Business analytics

B. Heatmap

A Heatmap uses color intensity to represent correlation or relationship strength among variables.

Example

Correlation matrix visualization.

Advantages

1. Easy interpretation.
2. Detects strong and weak correlations quickly.

Disadvantages

1. Difficult for extremely large datasets.

Applications

- Correlation analysis
- Feature selection

- Data science

C. 3D Scatter Plot

Represents relationships among three numerical variables in three-dimensional space.

Example

Relationship between:

- Age
- Income
- Spending score

Advantages

1. Better multidimensional visualization.

Disadvantages

1. Difficult interpretation sometimes.

Applications

- Business analytics
- Scientific visualization

D. Parallel Coordinates Plot

Visualizes multiple variables using parallel vertical axes.

Advantages

1. Useful for high-dimensional data.
2. Identifies clusters and trends.

Applications

- Big data analytics
- Pattern analysis

E. Bubble Chart

Bubble Chart extends scatter plots by adding bubble size as an additional variable.

Example

- X-axis → Sales
- Y-axis → Profit
- Bubble size → Customer count

Advantages

1. Represents multiple variables simultaneously.

2. Measures of Correlation for Numerical Attributes

Correlation measures the strength and direction of relationships among numerical variables.

In multivariate analysis, correlation identifies how variables influence one another.

Types of Correlation

A. Positive Correlation

Variables increase together.

Example

Experience and salary.

B. Negative Correlation

One variable increases while another decreases.

Example

Price and product demand.

C. Zero Correlation

No relationship exists between variables.

Correlation Matrix

A Correlation Matrix is a table showing correlation coefficients among multiple numerical variables.

Example

Variable	Age	Salary	Experience
Age	1.0	0.7	0.8
Salary	0.7	1.0	0.9
Experience	0.8	0.9	1.0

Importance

1. Detects relationships among variables
2. Supports feature selection
3. Helps reduce redundancy

Pearson Correlation Coefficient

Pearson correlation measures linear relationship between numerical variables.

Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Interpretation of Correlation Values

Correlation Value Interpretation

- +1Perfect positive correlation
- 0No correlation
- 1Perfect negative correlation

Advantages

1. Easy interpretation.
2. Detects linear relationships accurately.

Disadvantages

1. Sensitive to outliers.
2. Cannot detect non-linear relationships.

Spearman Rank Correlation

Spearman correlation measures relationships based on ranked values.

Formula

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Advantages

1. Works with non-linear relationships.
2. Less sensitive to outliers.

Applications of Correlation Analysis

1. Financial forecasting
2. Customer behavior analysis
3. Healthcare analytics
4. Machine learning feature selection
5. Risk analysis

Multicollinearity

Multicollinearity occurs when multiple independent variables are highly correlated with each other.

Problems Caused

1. Reduced model accuracy
2. Difficulty in interpretation
3. Unstable regression models

Detection Methods

- Correlation matrix
- Variance Inflation Factor (VIF)

Solution Techniques

1. Feature selection
2. PCA
3. Removing redundant variables

Advantages of Multivariate Analysis

1. Handles complex datasets
2. Detects hidden relationships
3. Improves predictive modeling
4. Supports advanced analytics
5. Better business insights

Disadvantages of Multivariate Analysis

1. High computational complexity
2. Difficult interpretation
3. Requires large datasets

Applications of Multivariate Analysis

1. Business Intelligence
2. Financial analysis
3. Healthcare analytics
4. Market research
5. Machine learning and AI

Difference Between Univariate, Bivariate, and Multivariate Analysis

Analysis Type	Variables Used	Purpose
Univariate	One	Analyze single variable
Bivariate	Two	Study relationships
Multivariate	More than two	Analyze complex interactions